From Prediction to Smarter Bank Targeting

An end-to-end data mining story | Portuguese bank marketing | Group 1

WHO should the bank call — BEFORE the call begins?

45,211 historical calls • 11.7% subscriptions • 4 algorithms • 1 operational decision

From data quality and exploration to leakage-aware modelling, ranking and deployment safeguards

The business problem

Every call consumes agent time. Only a minority convert.

TODAY

Broad calling lists

High effort

Many non-conversions

DATA MINING

Learn from historical outcomes

Rank likely subscribers

Support contact prioritization

BETTER DECISIONS

Contact within capacity

Measure real outcomes

Test before rollout

Goal: improve prioritization — not claim that prediction causes subscription.

The dataset in one glance

bank-full.csv | 16 original predictors + binary target

45,211
historical records

5,289
subscribed: YES

11.7%
positive class

Data audit

No null values or duplicate records. “unknown” categories retained as information.

Special value

pdays = −1 means no previous campaign contact; it is not a missing value.

What exploration revealed

Patterns help form hypotheses; they do not prove causal effects.

PAST CAMPAIGN

Previous success: 64.7% observed subscription vs 9.2% with no prior contact.

CUSTOMER SEGMENTS

Students: 28.7% | Retired: 22.8% observed subscription.

CONTACT INTENSITY

First contact: 14.6% | 11+ contacts: 3.9% (association only).

CALL DURATION

Over 10 minutes: 48.3% | Under 1 minute: 0.19%.

The most important modelling decision

A feature can predict extremely well — and still be unusable at decision time.

FULL-INFORMATION MODEL

Includes call duration. Useful as a diagnostic of predictive signal; duration is known only after the call.

PRE-CALL MODEL

Excludes duration. Strict final version also excludes campaign, which may reflect current-call activity.

Prediction quality ≠ operational usability

How we evaluated fairly

Common algorithm-comparison stage: original predictors, no duration, no imbalance correction.

5-FOLD CV

Stratified, shuffled folds

Seed = 42

Preprocessing fitted within training folds

MCC FIRST

Matthews correlation coefficient considers all four confusion-matrix outcomes; suited to imbalance.

SAME BASE

15 original predictors

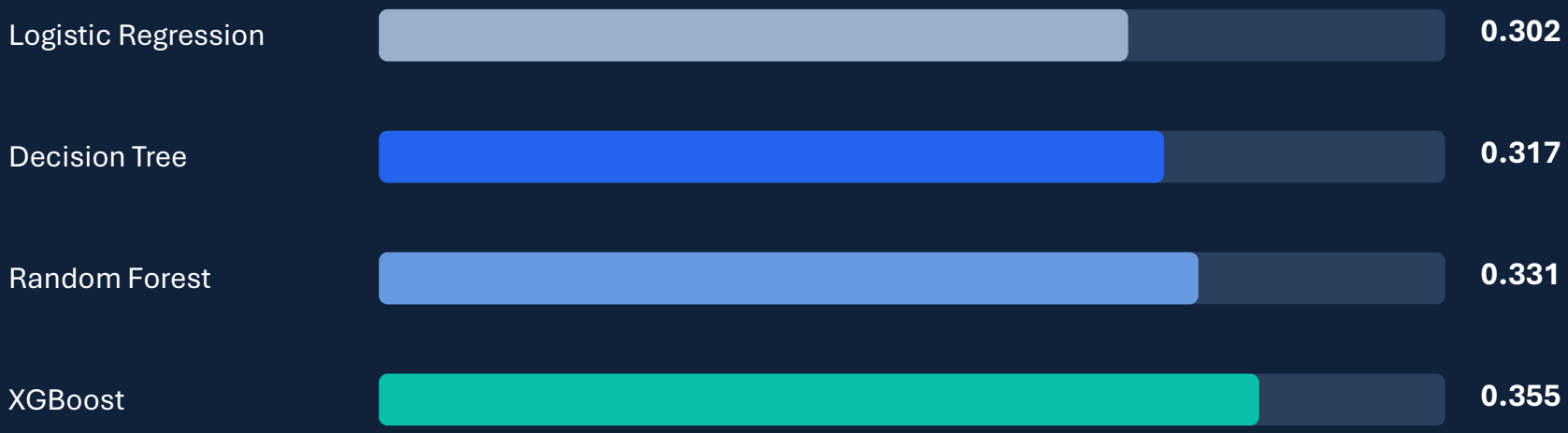
Threshold = 0.5

No feature engineering or rebalancing

A majority-class-only predictor achieves MCC = 0, despite 88.3% accuracy.

Four algorithms. One clear leader.

Tuned results; five-fold mean MCC; no duration, no imbalance treatment.



XGBoost led this controlled comparison. That does not prove it would lead under every possible tuning or weighting scheme.

Why model complexity mattered

Different algorithms exposed different trade-offs.

DECISION TREE

Tuning lifted MCC from 0.215 to 0.317 and reduced average leaves from ~4,763 to ~40.

RANDOM FOREST

Tuned MCC: 0.3305. Relative to its untreated baseline (0.3302), tuning alone barely changed MCC.

XGBOOST

Tuned MCC: 0.355. Selected for deeper feature, imbalance and operational tests.

Why call duration changes everything

Same model families evaluated with and without this post-call variable.



BLUE Without duration GREEN With duration (diagnostic only)

Feature engineering: complexity was not the answer

Separate XGBoost experiments: fixed settings, 5 folds × 3 seeds; not directly comparable to tuned Stage 1.

0.341

strict raw features

0.346

strict + engineered

0.348

interactions + selection

TESTED

Customer, balance, contact-history and timing transformations; interactions and selection.

DECISION

Gains were small and inconsistent. Retain the simpler strict raw predictors.

The class imbalance breakthrough

On tuned XGBoost without duration, moderate weighting improved the balance of errors.



Recall: 24.6% untreated → 52.2% at weight 4. Training-fold-only resampling avoids validation leakage.

The final model: realistic before the call

Tuned XGBoost + class weight 4 + strict original 14-feature set.

0.4211	51.6%	46.9%
MCC ± 0.0219	recall	precision

STRICT INPUT CONTRACT

No duration. No campaign. Five stratified folds, seed 42; default threshold 0.5. F1 = 0.4914.

From scores to a contact priority list

Out-of-fold historical ranking: each record scored by a model not trained on that record.

10%	43.85%	4.38×
customers selected	historical YES contained	historical lift

ANOTHER OPERATING POINT

Top 20% contained 62.19% of observed subscribers; 3.11× historical lift.

Not a promise of future conversions: these are retrospective out-of-fold estimates.

What the model is really learning

SHAP explains model associations, not causal drivers.

HIGH SIGNAL

Previous campaign outcome, month and contact channel contribute strongly to ranking.

HIDDEN RISK

Month, contact and day account for ~49% of mean absolute SHAP; these can proxy campaign phase.

A strong historical signal can be fragile when campaign conditions change.

The reality check: time changes the story

The original data are date-ordered; shuffled folds mix earlier and later campaign periods.

~0.80

shuffled-CV ROC AUC

~0.57

forward-time ROC AUC

DEPLOYMENT IMPLICATION

Historical top-decile lift ~4.38× in shuffled CV, but only ~1.3–1.5× in forward testing. Future-campaign validation is essential.

A practical path to deployment

Turn the model into a measured decision process — not an automatic calling policy.

01 Validate

Test on a later campaign.

02 Rank

Score only pre-call fields.

03 Pilot

Set contact volume by agent capacity.

04 Measure

Track lift, MCC, cost and fairness.

Takeaway: smarter targeting starts with realistic features, honest validation and operational feedback.

The project in one sentence

Better decisions require more than the highest model score.

AUDIT → EXPLORE → COMPARE → IMPROVE → VALIDATE → PILOT

Final historical MCC 0.4211 | Strict pre-call features | Time-aware deployment still required